

# Time Travel & Fail Safe



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# Section Introduction-Time Travel

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# What is Snowflake Time Travel ?

Snowflake Time Travel is a Snowflake feature that enables accessing data that has been changed, updated or deleted in the past.

## Time Travel Features & Benefits

- Time Travel helps in restoring data objects (tables, schema and databases) that might have been accidentally or intentionally deleted or updated.
- Time Travel helps in duplicating and backing up data from key events in the past.
- Eliminating the need to perform backups of data in Snowflake on a regular basis.
- Create clones of entire tables, schemas & databases at or before specific points in the past.
- Data retention is the number of days you can go in the past to get the older version of the data.
- Data retention can be set to be between 0 and 90 depending on the Snowflake Edition.
- By default the retention period is 1 day and it can be modified when creating the table or later with an ALTER statement.
- A table with a Data Retention period of 0 has Time Travel disabled and Time Travel operations will not work.
- Data retention period can be set at the database, schema or table level.
- Maximum amount of Time Travel permitted for Standard edition and higher is 1 day.
- Maximum amount of Time Travel permitted for Enterprise edition is 90 days.
- Data Retention period is determined by the “[Data Retention Policy](#)” of the company.
- Longer the Time Travel more storage space will be occupied to store the data and this will contribute to higher storage costs.

# What Fail-Safe?

Fail-Safe is a non-configurable 7-day period during which data deleted from Snowflake is temporarily saved and may be recoverable by Snowflake support. This period starts immediately after Time Travel retention period ends.

## Fail-Safe Features & Benefits

- Fail-Safe can be accessed by Snowflake support alone and cannot be accessed by the Data Engineer.
- Fail-Safe consumes storage and storage cost associated with Fail-Safe is billed to the Account.
- [ACCOUNT\\_USAGE.TABLE\\_STORAGE\\_METRICS](#) view has information on storage space consumed and you can review it there.
- Snowflake does not guarantee data recovery from Fail-Safe and is on best effort basis.
- Fail-Safe comes into play only after the Time Travel period has ended.

# Summary

- Snowflake Time Travel is a feature of Snowflake that enables accessing data that has been changed, updated or deleted in the past.
- Time Travel helps in restoring data-related objects (tables, schema and databases) that might have been accidentally or intentionally deleted or updated.
- Time Travel period or data retention is set by the parameter `DATA_RETENTION_TIME_IN_DAYS` which determines how long the data will be stored in Time Travel.
- By default the retention period is 1 day and can be modified when creating the table or later with an ALTER statement
- Data retention period (`DATA_RETENTION_TIME_IN_DAYS`) can be set to be between 0 and 90 depending on the Snowflake Edition.
- Data retention period (`DATA_RETENTION_TIME_IN_DAYS`) can be set at the database, schema or table level.
- Data retention period (`DATA_RETENTION_TIME_IN_DAYS`) set at the lowest level takes precedence over that set at a higher level for that reason data retention at the table will have higher priority than schema and data retention set at schema will have higher priority than database.
- Time Travel eliminates the need to perform backups of data in Snowflake.
- Time Travel has simplified the process of recovery of data in Snowflake saving time and effort for Data Engineers

# Summary

- There are 3 Time Travel options you can use with a select statement
  - `AT(TIMESTAMP => '2025-01-27 12:08:32.338 -0700':TIMESTAMP_LTZ)`
  - `BEFORE(STATEMENT => '8e5d0ca9-005e-44e6-b858-a8f5b37c5726')`
  - `AT(OFFSET => -60*5)`
- Use the `UNDROP` statement to recover dropped tables, schemas and databases.
- Time Travel can be turned off by setting `DATA_RETENTION_TIME_IN_DAYS=0`
- Fail-Safe is a non-configurable 7-day period during which data deleted from Snowflake is temporarily saved and may be recoverable by Snowflake support.
- Fail-Safe period starts immediately after Time Travel retention period ends.
- Fail-Safe can be accessed by Snowflake support alone and cannot be accessed by the Data Engineer.
- Fail-Safe consumes storage and costs associated with Fail-Safe will be billed to the customer.
- Snowflake view `ACCOUNT_USAGE.TABLE_STORAGE_METRICS` has information on storage space consumed and you can review Fail-Safe storage consumed.



Thank you



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